

SymPlan: Enhancing Neurosymbolic Mathematical Reasoning in Compact Language Models via Representation and Planning

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Abstract. Executable symbolic programming enables language models to overcome calculation errors in multi-step mathematical reasoning. While frontier proprietary models excel at this task by synthesizing interpretation, planning, and code in a single unconstrained pass, compact small language models (SLMs, $\leq 8\text{B}$ parameters) under 4-bit quantization collapse under this monolithic burden. Due to limited capacity, compact models struggle to concurrently parse problem semantics, plan algebraic steps, and emit formal code, leading to semantic goal drift, omitted domain boundaries, and repair regressions. To bridge this capability gap without parameter scaling, we introduce **SymPlan**, an inspectable neurosymbolic framework designed for resource-constrained reasoning. Rather than relying on monolithic generation, SymPlan establishes an immutable *target anchor* that fixes the queried unknown and its formal output signature prior to code generation, stabilizing downstream SymPy synthesis and preventing regressive overwrites during iterative debugging. Evaluated across six open-weight models (3B–8B) on the MATH-500 benchmark under 4-bit quantization, SymPlan consistently outperforms monolithic SymCode by +3.60% to +5.20% (averaging +4.20%) and Chain-of-Thought by +12.64%, while elevating the execution success rate to 89.90% without fine-tuning.

Keywords: Neurosymbolic AI · Small Language Models · Mathematical Reasoning · SymPy Code Generation · Algorithmic Planning.

1 Introduction

Mathematical reasoning in language models has advanced through Chain-of-Thought (CoT) prompting [15] and structured search [14,16,13]. Nevertheless, autoregressive generation often produces arithmetic errors, omitted steps, and unfaithful derivations that resist verification [8,7]. Program-aided methods address these weaknesses by delegating computation to an external interpreter: PAL [4] and PoT [1] run numerical code in Python, MathCoder [12] and ToRA [5] blend tool calls with rationales, and SymCode [10] uses SymPy [9] for exact algebra.

However, program execution does not guarantee mathematical correctness: neurosymbolic frameworks fundamentally depend on whether the formalization faithfully represents the problem [8,11]. Most existing methods generate interpretation, planning, and code in a single prompt [4,10]. Although frontier models often manage this combined task, compact small language models ($\leq 8\text{B}$ parameters) under 4-bit quantization [3] degrade severely on MATH-500 [7,6]. In our experiments, these models exhibit four failure patterns: (1) *Semantic misformulation*: the code solves a distorted or incomplete equation; (2) *Target-anchor drift*: the program drifts from the queried objective and type signature, returning lists for cardinality or floats for fractions; (3) *Domain boundary omission*: extraneous roots are accepted without filtering against domain bounds; and (4) *Silent repair regression*: self-debugging yields valid scripts that replace previously correct answers with incorrect ones.

To address these issues, we present **SymPlan**, an inspectable neurosymbolic framework for compact models. SymPlan structures problem solving into three stages: first, a *Fused Target-Anchoring Phase* establishes an explicit target anchor with the queried variable, formal output signature, and algorithmic roadmap (≤ 256 tokens); second, a *Target-Anchored Code Generator* translates this specification into self-contained SymPy code enforcing domain boundaries and exact arithmetic; third, a *Targeted Debugging Loop* pairs tracebacks and AST diagnostics with an *Anti-Regression Selection* mechanism to preserve valid intermediate solutions.

Our primary contributions are: (1) characterizing formulation, target-anchor drift, and repair failure modes across six compact models on MATH-500; (2) introducing SymPlan, uniting fused target anchoring, specification-guided SymPy synthesis, and anti-regression ranking; and (3) demonstrating that SymPlan outperforms SymCode by +3.60% to +5.20% (and CoT by +12.64%) across six quantized SLMs, increasing ESR to 89.90% without fine-tuning.

2 Related Work

Program-Aided Reasoning. PAL [4] and PoT [1] offload numerical steps to Python, MathCoder [12] and ToRA [5] combine derivations with code execution, and SymCode [10] utilizes SymPy [9] for exact algebra. However, monolithic generation overburdens compact models: if the model misinterprets equations or omits boundary conditions, the interpreter returns mathematically invalid answers.

Formalization and Code Repair. Faithful CoT [8] separates reasoning from execution, and Logic-LM [11] refines symbolic formulations via solver feedback. For code repair, Self-Debugging [2] iteratively revises scripts via interpreter feedback. On compact models, unanchored repair frequently induces code cycling or silent regressions. SymPlan grounds the repair process: the immutable target anchor preserves mathematical intent and output signature, while the attempt log prevents regressive overwrites.

Planning in SLMs. Structured prompting improves reasoning in compact architectures, including Self-consistency [14], Least-to-Most [16], and Plan-and-Solve [13]. SymPlan builds on these insights by establishing an explicit target anchor and structured execution steps designed for quantized models.

3 Method

3.1 Framework Formulation

Given a mathematical problem q , monolithic symbolic approaches [10] synthesize and execute code in a single unconstrained step: $C_{\text{base}} \sim \mathcal{M}(\cdot \mid q)$, $\hat{y} = \text{Exec}(C_{\text{base}})$. On compact models, this joint generation causes semantic drift and execution crashes. SymPlan structures problem solving through three target-anchored phases:

$$\mathcal{T} = (\tau, \Sigma, \mathcal{P}) = \text{FusedPlan}_{\mathcal{M}}(q), \quad (1)$$

$$C^{(0)} = \text{CodeGen}_{\mathcal{M}}(q, \mathcal{T}), \quad \mathcal{H} = \text{ExecuteAndRepair}_{\mathcal{M}}(C^{(0)}, q, \mathcal{T}), \quad (2)$$

$$\hat{y} = \text{SelectBest}(\mathcal{H}, \Sigma), \quad (3)$$

where τ denotes the queried target unknown, Σ is the formal output signature, $\mathcal{P}$ represents the ordered computational plan, and $\mathcal{H}$ records the execution history. Like a holding anchor stabilizing team structure, establishing $\mathcal{T}$ prior to code generation anchors mathematical intent throughout subsequent execution and debugging.

3.2 Fused Target Anchoring and Algorithmic Planning

Multi-turn prompting pipelines often double autoregressive latency and cause context drift in compact models. To avoid this overhead, SymPlan extracts the target anchor and plan tuple $\mathcal{T} = (\tau, \Sigma, \mathcal{P})$ in a single bounded generation (≤ 256 tokens): (1) **Target Specification (τ):** the formal quantity queried, such as the positive real roots of an equation; (2) **Formal Output Signature (Σ):** a discrete type signature $\Sigma \in \{\text{numeric}, \text{text}, \text{tuple}, \text{set}, \text{symbolic}\}$, where queries asking “how many” assign $\Sigma = \text{numeric}$ to return set cardinality rather than enumerating values; and (3) **Computational Plan ($\mathcal{P}$):** an ordered sequence of symbolic operations $(p_1, \dots, p_m)$ defining symbols, equations, transformations, and domain boundary checks.

3.3 Target-Anchored SymPy Code Generation

Given the target anchor and plan $(\tau, \Sigma, \mathcal{P})$, generator $\text{CodeGen}_{\mathcal{M}}$ produces a self-contained SymPy script following four rules: (1) *Exact Arithmetic:* Rationals use `sp.Rational(p, q)` or `sp.S(p)/q`; float evaluations via `evalf()` are excluded unless decimals are requested. (2) *Domain Filtering:* Roots from `sp.solve` are

Table 1. Cross-model accuracy (%) and Execution Success Rate (ESR %) on MATH-500 across six compact models (3B–8B). Best accuracy is in **bold**.

Model	Size	Direct	CoT	SymCode	SymPlan	ESR (SC)	ESR (Ours)
Qwen2.5-3B-Instruct	3B	17.00	39.80	37.40	41.20	80.20%	87.20%
Qwen3-4B	4B	36.00	45.40	56.40	61.60	81.60%	90.20%
Qwen2.5-7B-Instruct	7B	25.40	45.60	54.00	57.60	88.40%	93.80%
Qwen2.5-Coder-7B-Instruct	7B	22.00	38.20	57.00	61.40	89.40%	92.40%
Llama-3.1-8B-Instruct	8B	23.20	37.60	37.60	41.80	76.60%	84.60%
Qwen3-8B	8B	36.20	45.00	59.80	63.80	86.80%	91.20%
Average (6 Models)	All	26.63	41.93	50.37	54.57	83.83%	89.90%

filtered against domain bounds ($x \in \mathbb{R}^+$, non-zero denominators). (3) *Simplification*: Expressions are simplified via `sp.together()` or `sp.cancel()`. (4) *Signature Conformance*: The script emits a single boxed $\text{\LaTeX}$ answer strictly conforming to signature Σ .

```
import sympy as sp; x = sp.Symbol('x', real=True); eq = sp.Eq(sp.sqrt(x), x - 2)
cands = sp.solve((x - 2)**2 - x, x); valid = [s for s in cands if s >= 2 and eq.lhs.subs(x,
s) == eq.rhs.subs(x, s)]
print(r"\boxed{%s}" % sp.latex(valid[0]))
```

3.4 Targeted Debugging with Anti-Regression Selection

Scripts execute in an isolated sandbox with timeout $\tau = 15\text{s}$ and retry limit $K = 2$, incorporating three safeguards: **AST Diagnostics**: Tracebacks $e^{(i)}$ are paired with static AST diagnostics (`lint_sympy_code`) and candidate verification $v^{(i)}$. **Anti-Cycling Repair**: If $C^{(i)} \equiv C^{(i-1)}$, an explicit constraint forces an alternative implementation strategy. **Anti-Regression Ranking**: Every attempt is logged into history $\mathcal{H} = \{r_0, \dots, r_k\}$ and scored via a lexicographic ranking tuple $\text{Rank}(r) = \langle \mathbb{I}_{\text{exec}}(r), \mathbb{I}_{\text{valid}}(r), \mathbb{I}_{\text{verif}}(r), -\text{attempt}(r) \rangle$, where $\mathbb{I}_{\text{exec}}$ indicates zero execution errors, $\mathbb{I}_{\text{valid}}$ denotes valid boxed extraction, $\mathbb{I}_{\text{verif}}$ indicates candidate verification pass, and $-\text{attempt}(r)$ prioritizes earlier valid runs over speculative later mutations. The optimal record is $r^* = \arg \max_{r \in \mathcal{H}} \text{Rank}(r)$. Multi-tier fallback handles unparseable outputs via regex scanning and type normalization.

4 Experiments and Results

Setup. We evaluate on the full MATH-500 benchmark [7,6] (7 subjects, 5 difficulty levels) across six open-weight models: Qwen2.5-3B-Instruct, Qwen3-4B, Qwen2.5-7B-Instruct, Qwen2.5-Coder-7B-Instruct, Llama-3.1-8B-Instruct, and Qwen3-8B. All models run under 4-bit NormalFloat4 (NF4) quantization [3] with greedy decoding (`temp = 0.0`, `max_tokens = 1024`). We benchmark against Direct zero-shot generation, Chain-of-Thought (CoT) [15], and monolithic SymCode [10], evaluating Accuracy (% EM) and Execution Success Rate (ESR).

Main Results. Table 1 reports results across the evaluated models. SymPlan consistently outperforms monolithic SymCode by +3.60% to +5.20% (averaging

Table 2. Fine-grained accuracy (%) across difficulty levels and subjects on MATH-500 for Qwen2.5-Coder-7B and Qwen3-8B.

Category	Count	Qwen2.5-Coder-7B		Qwen3-8B	
		SymCode	SymPlan	SymCode	SymPlan
<i>Difficulty Levels</i>					
Level 1–2	133	78.95	84.96	75.19	81.20
Level 3	105	61.90	73.33	63.81	66.67
Level 4–5	262	43.89	44.66	50.38	53.82
<i>Mathematical Subjects</i>					
Algebra	124	66.94	76.61	71.77	75.81
Counting & Probability	38	52.63	57.89	44.74	57.89
Geometry	41	31.71	34.15	46.34	51.22
Intermediate Algebra	97	51.55	48.45	50.52	51.55
Number Theory	62	66.13	70.97	62.90	72.58
Prealgebra	82	69.51	75.61	73.17	73.17
Precalculus	56	37.50	41.07	46.43	48.21

Table 3. Failure mode comparison between SymCode and SymPlan on MATH-500.

Failure Category	SymCode	SymPlan	Net Red.	Rel. Red.
Execution Failures (Syntax/Crash)	485 (16.2%)	303 (10.1%)	-182	-37.5%
Format / Type Mismatches	515 (17.2%)	394 (13.1%)	-121	-23.5%
Semantic Misformulations	974 (32.5%)	969 (32.3%)	-5	-0.5%
Total Incorrect Solutions	1489 (49.6%)	1363 (45.4%)	-126	-8.5%

+4.20%) and CoT by +12.64% (reaching +23.20% on Qwen2.5-Coder-7B). Average ESR increases from 83.83% to 89.90% (+6.07%), lowering execution failures by 37.5%. In inference cost, SymPlan generates only 22.6% more tokens than SymCode (392.9 vs. 320.6 tokens/problem), remaining substantially shorter than CoT (423.7 tokens).

Stratified Analysis and Ablation. Table 2 indicates that gains concentrate in intermediate problems (Level 3: +11.43% on Coder-7B). By subject, improvements focus on *Algebra* (+9.67%), *Counting & Prob.* (+13.15%), and *Number Theory* (+9.68%), where integer constraints are easily omitted by monolithic baselines. In component ablations, target anchoring (+1.60% to +2.40%) and algorithmic planning (+2.40% to +3.80%) each improve accuracy independently, while combining them with anti-regression ranking achieves the highest accuracy (+3.60% to +5.20%).

Error Diagnostics and Mechanistic Insights. As detailed in Table 3, SymPlan eliminates 126 errors across 3,000 evaluations (−8.5% relative drop). Safe SymPy templates reduce runtime crashes by 37.5%, while formal output signatures lower format and type mismatches by 23.5%. In multi-condition algebra problems, unconstrained generation frequently overlooks intermediate boundary conditions during code synthesis. By contrast, the target anchor establishes an explicit algebraic contract registering all evaluation points and domain bounds beforehand, preventing constraint omission and enabling exact symbolic recon-

struction. Conversely, semantic misformulations remain flat ($974 \rightarrow 969, -0.5\%$), indicating that neurosymbolic scaffolding resolves interface-level discrepancies while relying on base-model capabilities for non-constructive mathematical formulation.

5 Conclusion

We presented SymPlan, an inspectable neurosymbolic framework for mathematical reasoning in compact models. By decoupling problem solving into fused target anchoring, specification-guided SymPy synthesis, and targeted debugging with anti-regression ranking, SymPlan resolves semantic misformulation, domain boundary omission, and repair regressions. Across six open-weight models (3B–8B) on MATH-500 under 4-bit quantization, SymPlan outperforms monolithic SymCode by +3.60% to +5.20% and CoT by +12.64%, while raising execution success to 89.90% without parameter updates. Future work will extend SymPlan to geometric diagrams and interactive theorem proving.

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